POST

# Latest news . . .

May 4, 2020

AI & the Environment

Volume IV, May 4, 2020

---“Protecting the environment. Simpler. Faster.”

—Ceres AI

Environmental AI

Updates and recent headlines on artificial intelligence and the environment.

From the Editor

Regulations slow … algorithms fast.

---“Humans convert data to language, numbers, and images … all inputs that are inherently slower at communicating information to machines than straight-up data. If we leave environmental data in its original format as much as possible with algorithms that adjust to meet environmental protection goals—environmental compliance becomes much faster and simpler.”-Jed Anderson, Creator, Ceres AI

MIT creates Automated AI “Curiosity” Algorithms

MIT researchers in the last few weeks created a “meta-learning” algorithm that generates its own exploration algorithms.

Curiosity is a main driver of intelligence. AI can now begin to learn what a cow looks like because it's curious what a cow looks like … not just because it was forced by a human to look at 1,000 pictures of a cow.

Read More. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yMQu2KX1UjIhAHEEfRL9ptthLduKUQ-4iS9flc1WkRJu24SK4Sr_2pz8MVtA482SidJV8Ur7FdX8Uvi_K6PixF-EbdWxP5ueAHt7c4NnuzXB8L3s-CrePYbvtCaefnJ4umITnkMOx4hp3uhLOEVXJIWnpg33Xe4M1&c=&ch=)

Ceres Expands "AI Earth"

- Terabytes of EPA data
- Terabytes of TCEQ data
- Increased satellite resolution
- Pipelines and oil wells
- Bounding-box
- 150 environmental program areas

---"AI Earth unites regulatory data with geo-spatial data to deliver new compliance and business insights."-Bradford Tuckfield, PhD, Chief Technology Officer, Ceres AI

For more information. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFlUAjLQiDf2U8Tdrw1EC_EP1E35FslHxKgetlvdEOCRPS1z8mgdb0Q5UHEYUpw39Z-OdH-d7auQX_RxqFB67HDsRgy0bhov2gg==&c=&ch=)

Google Announces AutoML Zero

Google has announced a new automatic machine learning tool called “AutoML Zero”. “Human-designed components bias the search results in favor of human-designed algorithms, possibly reducing the innovation potential of AutoML,” […] “Innovation is also limited by having fewer options: you cannot discover what you cannot search for.”

“AutoML-Zero is unique because it uses simple mathematical concepts to generate algorithms from scratch and then it selects the best ones and mutates them through a process that’s similar to Darwinian evolution.”

Read More. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yMDi-MtH07TnA-PE0AEdrEZpJsbh14kv1jhg--S4hMcuQstLmdWkJO3N-8RBjTk9F3VXRxNslKAGbVsTXvVrMNxC1-VJ8AMAxM9pyO20uMnM-03CCEP_KcVkop3gjZgdnz6rYqYOAxuCQcizT4972VU2qUlY9x3S0tjpPT02vz2qIjzpTwTNdOeW1PxQszfH8rZGpeVgPGNA=&c=&ch=)

A.I. Environmental Due Diligence

Ceres AI is making environmental due diligence much simpler and faster for companies, bankers, and others interested in quickly reviewing external data about a particular site or company.

Read more. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFkyELQxuWLZFvPFSaFGxpq4B03QQ9BmRF0f_bT5YuPaHv--aAafp1FM70exWDLVlKHxwM2bc49qQT069PImA3lb6ZrucWneieKKiGYww4LjtxwCU9icZgt_uVbMkA4DZuYYi_WRPv8Su7lb5F_tGOEIKXeZqM2kkv2sJc5z9NJ9HxQWkJhznRhM=&c=&ch=)

Quantum Leaps in Quantum Networking and Computing

Major milestones toward the quantum internet and quantum computing in the past few months:

- Argonne Laboratory 30 mile Quantum Internet under Chicago (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yzWewar32stoVy7GT4GVxFRr3x2449awqUx7x0JqTvBP47krjymJX6qQ0YsrzLf_0ZXr0YAM-kDmHSm2eWIvHWppFtGeUyplfULTCX5MneLbPpSxCyduRVUM8iS1mRhf0-RdtQqSDoiod9VWovtDISttYrzbsyTrh&c=&ch=)
- Chip-to-chip Data Teleportation (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yV43h1JuZHq_0wUV0I7wywcJphbvfhsBRjlehOGhQBx27VVyQ6IMxI_KE1HKKKyFol8Ojh-qqWdoOcHDsOj3S_XR-vimzy2u-2vLIP_QepGpwDK8CDIbFK7lOD6V3hOaQFlNrx641YB8b5wvx3yN8nKX7BsOzZyRlI8pneDVLSzk=&c=&ch=)
- Hot Temp Quantum Computing (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yTxlZilVy4d3nFWCw2K01VGv0Bi8emJQArttqjpg0jYOxwguscRtqLl28q8I6si5o4deie21kLZ6r4TdzrmEUL7Qh4ZLZzKigplqSYiL1AUCCj1lYZPhmC7rMnGC0IUutBasNKkEGhRXIXU2CWX3SUw==&c=&ch=)
- Quantum Entanglement for GPS —unparalleled accuracy (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yQXw4bLtnARnuC6PmntXJOHgRxh8HmCKqMqJzsn51tYDb_r0RFA0fymERRwMxszrtYu_ZbvgSJwdrvM1gDetwpna_S_GR2mT4Sp617ozPPsJdU_7m1be854zJBtDwygwg1-3N8G8N0a8=&c=&ch=)

—“Quantum networking doesn’t speed the car up, it removes the needs for roads entirely.”—NextWeb

—“A quantum internet will be the platform of a quantum ecosystem, where computers, networks, and sensors exchange information in a fundamentally new manner where sensing, communication, and computing literally work together as one entity.”— David Awschalom, Professor of of Molecular Engineering, University of Chicago.

“—As Richard Feynman said, ‘nature is not a classical system … it’s a quantum system’. Therefore, once we have a quantum system to measure, calculate, and communicate a quantum system—and artificial intelligence to help us manage this intensive stream of data—progress in environmental protection will become very fast and simple. Technology in the 1st industrial revolution began hurting the earth … and ironically, in the greatest twists of planetary fate … its technology in the 4th industrial revolution that might be the only reason why it ever survived.”—Jed Anderson, Creator, Ceres AI

Next Steps for “AI Earth”

AI earth will begin to allow users to explore different areas of Texas and ask Ceres AI various environmental questions about that particular area.

Read more. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6y-MibukHZY0brlXu0NmlY2_6O3kqCHf8IJy_5QyDChjQ4Rt2GQN6XNMVmSqr89Hu9Pdxx0Ab58xas_Q-Go6XiRA==&c=&ch=)

Ceres AI will also begin to provide more virtual environmental management and compliance solutions.

See video. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFthm1gbgmhVj3U9Ude6YGnb_0BOOznwCbCwEAotNWs0IctgvbIk_VSw8-2qrKF2A1_OPZnav8eub8vCpIhV9eMkSc9mznHHMuQ==&c=&ch=)

Elon Musk's "SpaceX" and Jeff Bezos' "Blue Origin"

Why are both billionaires enamored with space? Why are both along with Richard Branson in a race to put as many satellites into low-orbit as possible?

Many, many reasons . (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yFm-BJrw0SrwHefCHN6g-xvYXpzhfMDR1UI6P3dxXHcFcl8Ew1mkBfEPlJUJLw6VYOsrS4uXN9kebmp2EhXnuS9RRrE4dlKAAPyBbKeBHna67K0FRBGIZeQ4eNGT9cZRzshWX32kgD8fk2hc1KVfFMXtEmBqpI-7XnVHsCnJeLL8ZRnrVfXvNKj4GgAZEFnUe&c=&ch=) . . but here's one. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yFm-BJrw0SrwHefCHN6g-xvYXpzhfMDR1UI6P3dxXHcFcl8Ew1mkBfEPlJUJLw6VYOsrS4uXN9kebmp2EhXnuS9RRrE4dlKAAPyBbKeBHna67K0FRBGIZeQ4eNGT9cZRzshWX32kgD8fk2hc1KVfFMXtEmBqpI-7XnVHsCnJeLL8ZRnrVfXvNKj4GgAZEFnUe&c=&ch=) Here's a recent MIT article (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yche3N9N2c6se8GHBP9xBEStC9g1KdQvCD-Q1dqsNcMpcoYcXfde8FRviJQLPthO3RHJEjQUf90hqOyllxQ_G6g2k3hNpM-O0ebHcOZYvf1d18nl89IOUqu_EmsDj4AMPC_Dy4O8bQkvMcrmtSRyVTl3eBarVdErXEHEljCJFOCKP4HO4eb_mDh_k9H8ELKV2JLayydSN0s0=&c=&ch=) on why the quantum internet will likely be built in space. "The best way to distribute quantum entanglement around the globe is via a massive constellation of orbiting satellites, physicists say." The reason is the benefits of handling sensitive quantum entangled particles in the vacuum of space (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFqSvTL-s4s6yjajw8CYCQ2ShqxjgY55nuFxvkeUvZ3N1f9GWLNBub1apdpCp3lrb9ji5t4scL-vLKrsYIFsPjk4OB2Ys7Se4tehfGFzyMkFVIYuRkLjZmV9e5_Ta_bKw31ieRYAFTo_QMElU8z5rWCVp2WJA16JCD5xk64zb_ycqp9yux2Wlac_43HDQVBtboBxNTCinCLdeEqD4D2fttYMJesG9A1R00QHy5Dq_S5t36XZJdZ6d0DE=&c=&ch=).

Ceres launches "The Generator"

The Generator sifts through facility and equipment data to show potential emission reductions and credit opportunities.

For Ceres access. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFg07_4D4zExyQjQr15G0lIqXNJQTSJiDQhSzVM3_FDVNzNmBfBqpsXtNKQgkQ7HyBtxxRjYpz4o7LTpq-CPlkkef19r_0kfxSCNaDXGNmHzO5rrg70UkfSI=&c=&ch=)

Ceres AI's Search … "Google" for environmental

Ceres AI provides unparalleled access to environmental data.

The databases and AI algorithms in this system continually grow and expand.

More data. More knowledge. More power.

Read more or to purchase. (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFra4kONbHp9j9gKHrHVNyCU2jb1EVaZZbs1OzvYLE7ckQcvQKo25CaUgkz-VdL27z1Q66EFuwROva3jKgGXjzN1JybrCrWHlpQ==&c=&ch=)

Information. Data. Power. Speed.

This is an advertisement for Ceres AI

For more information on Ceres AI (http://r20.rs6.net/tn.jsp?f=001kw1MljRLTti4NFI1XQuz9CsqMuCbWP_MycY0G3mSswyrthMp0jtAFra4kONbHp9j9gKHrHVNyCU2jb1EVaZZbs1OzvYLE7ckQcvQKo25CaUgkz-VdL27z1Q66EFuwROva3jKgGXjzN1JybrCrWHlpQ==&c=&ch=)

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